

Design and Implementation of CSI Controller IP



Register file interfaced with APB

Transmitter Register File

TX Register Summary

Offset	Register Name	Description
0x00	TX_CTRL	Transmitter Control
0x04	TX_CFG	Transmitter Configuration

TX_CTRL Register (Offset 0x00)

Bit	Name	Access	Description
0	TX_EN	R/W	Enables CSI-2 transmitter
1	SW_RST	R/W	Software reset (self-clearing)
31:2	RSV	–	Reserved

TX_CFG Register (Offset 0x04)

Bit	Name	Access	Description
0	SCRAM_ENB	R/W	Scramble Enable
1	CRC_ENB	R/W	CRC Enable
31:2	RSV	–	Reserved

Receiver Register File

RX Register Summary

Offset	Register Name	Description
0x00	RX_CTRL	Receiver Control
0x04	RX_CFG	Receiver Configuration
0x0C	RX_ERR	Receiver Error Status

RX_CTRL Register (Offset 0x00)

Bit	Name	Access	Description
0	RX_EN	R/W	Enables CSI-2 receiver
1	SW_RST	R/W	Software reset (self-clearing)
31:2	RSV	—	Reserved

RX_CFG Register (Offset 0x04)

Bit	Name	Access	Description
0	RX_SCRAM_ENB	R/W	Scramble Enable
1	RX_CRC_ENB	R/W	CRC Enable
31:4	RSV	—	Reserved

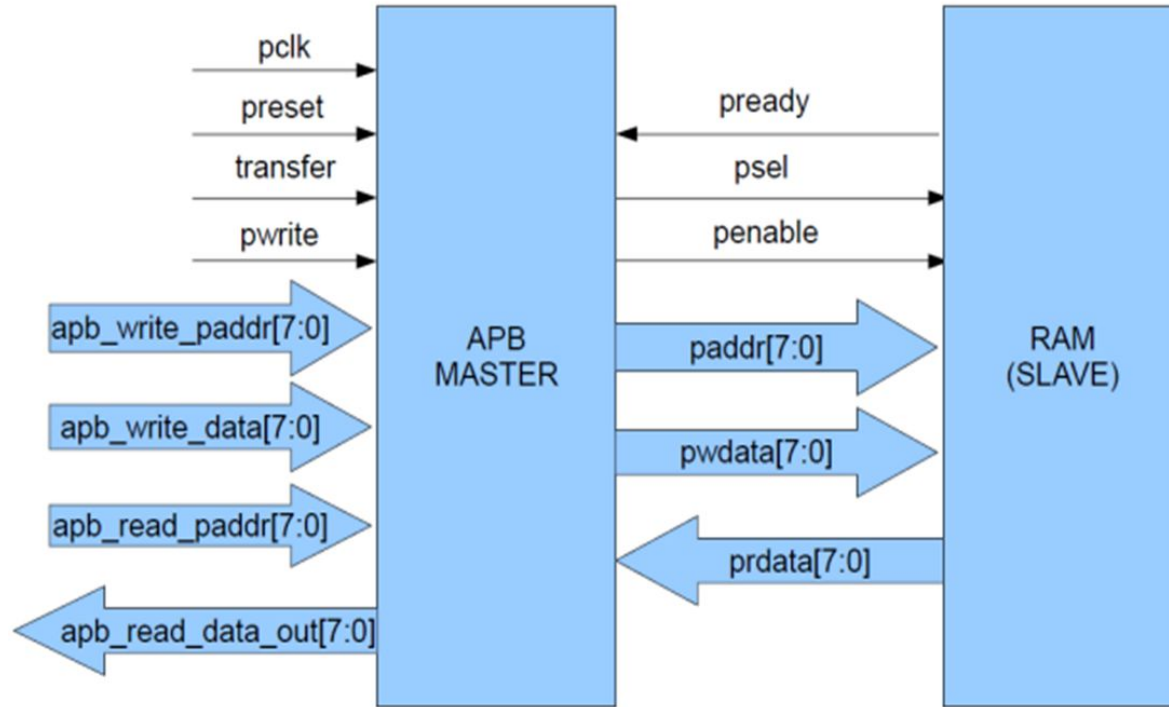
RX_ERR Register (Offset 0x0C)

Bit	Name	Access	Description
0	CRC_ERR	R	Payload CRC error
1	ECC_ERR	R	Header ECC error
2	ECC_ERR	R	Corrected ECC error
31:3	RSV	—	Reserved



APB Protocol

Advanced Peripheral Bus



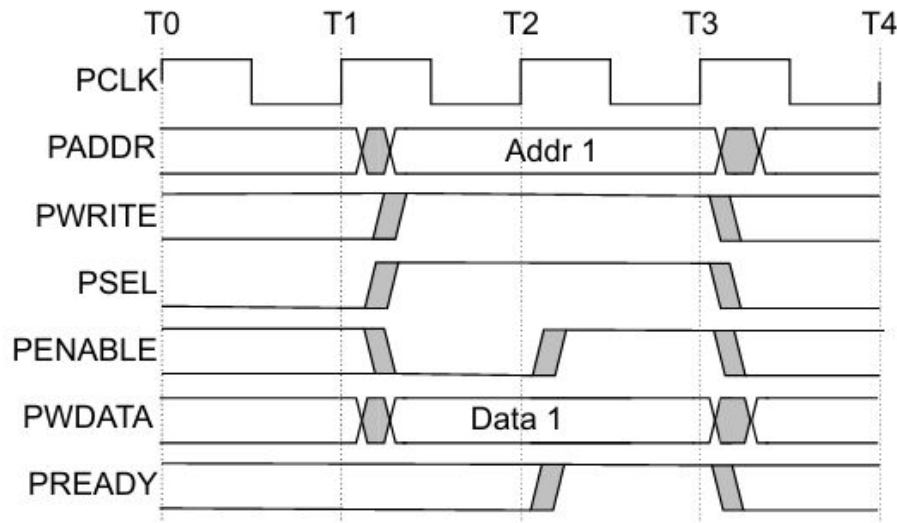
Ports Names

Advanced Peripheral Bus (APB) is a **single-master, multi-slave** bus protocol.

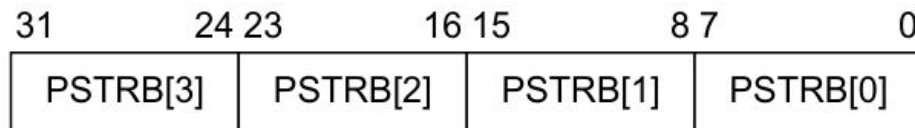
Signal	Signal Name
PCLK	Clock signal
PRESETn	Reset signal
PADDR	32 bit address bus
PSELx	Select signal
PENABLE	Enable signal
PWRITE	Direction signal
PWDATA	32 bit Write Data bus
PREADY	Ready signal
PRDATA	32 bits read data bus

Write Transfers

- The write transfer starts with the address, write data, write signal and select signal all changing after the rising edge of the clock.
- The first clock cycle of the transfer is called the **Setup** phase.
- After the following clock edge the enable signal is asserted, PENABLE, and this indicates that the **Access** phase is taking place.



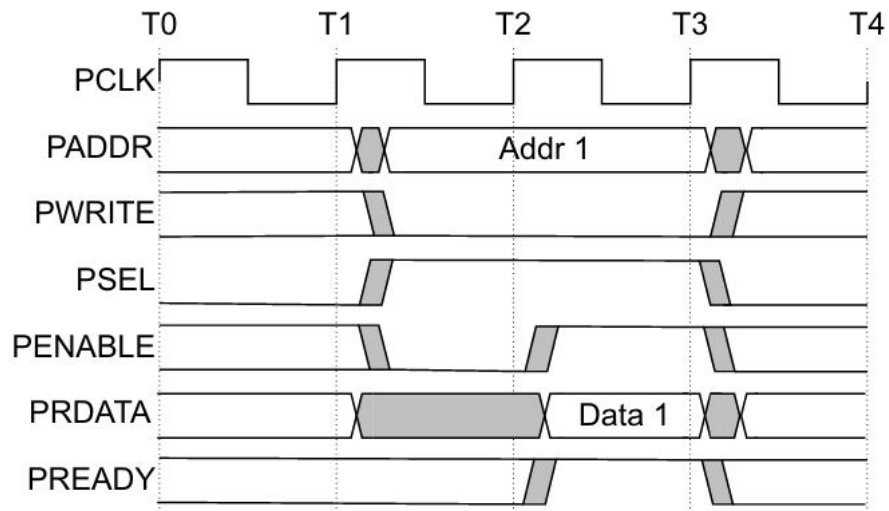
Write strobes



PSTRB enables sparse data transfer on the write data bus. Each PSTRB corresponds to 1 byte of the write data bus. When asserted HIGH, PSTRB indicates that the corresponding byte lane of the write data bus contains valid information.

For read transfers, the Requester must drive all bits of PSTRB LOW.

Read Transfers



Operation of APB

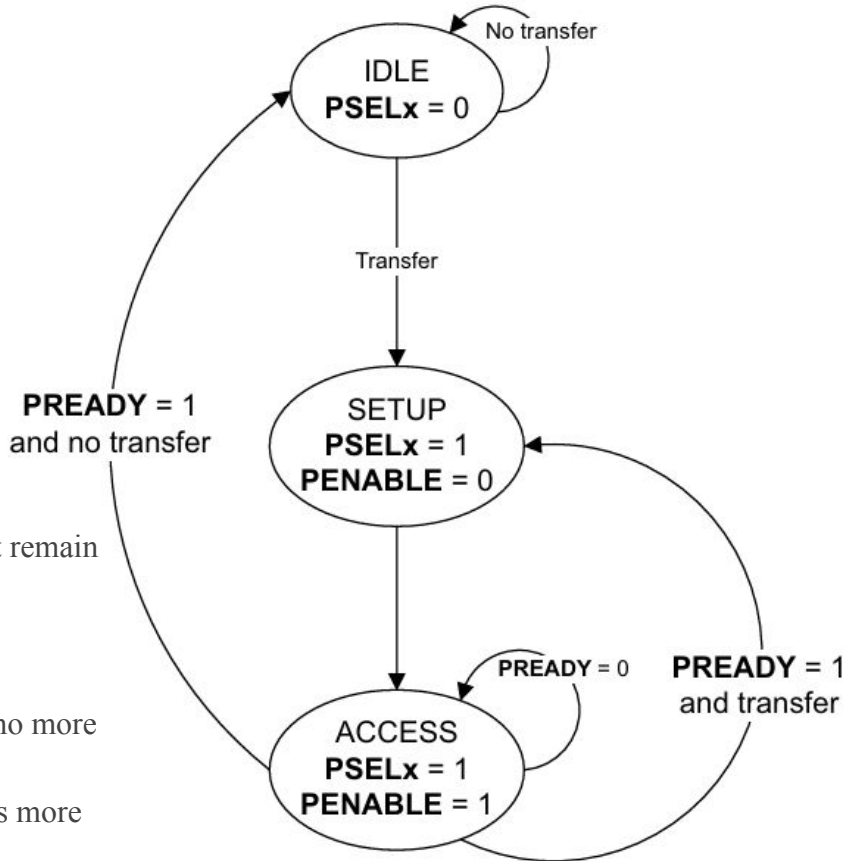
IDLE: This is the default state of the APB.

SETUP:

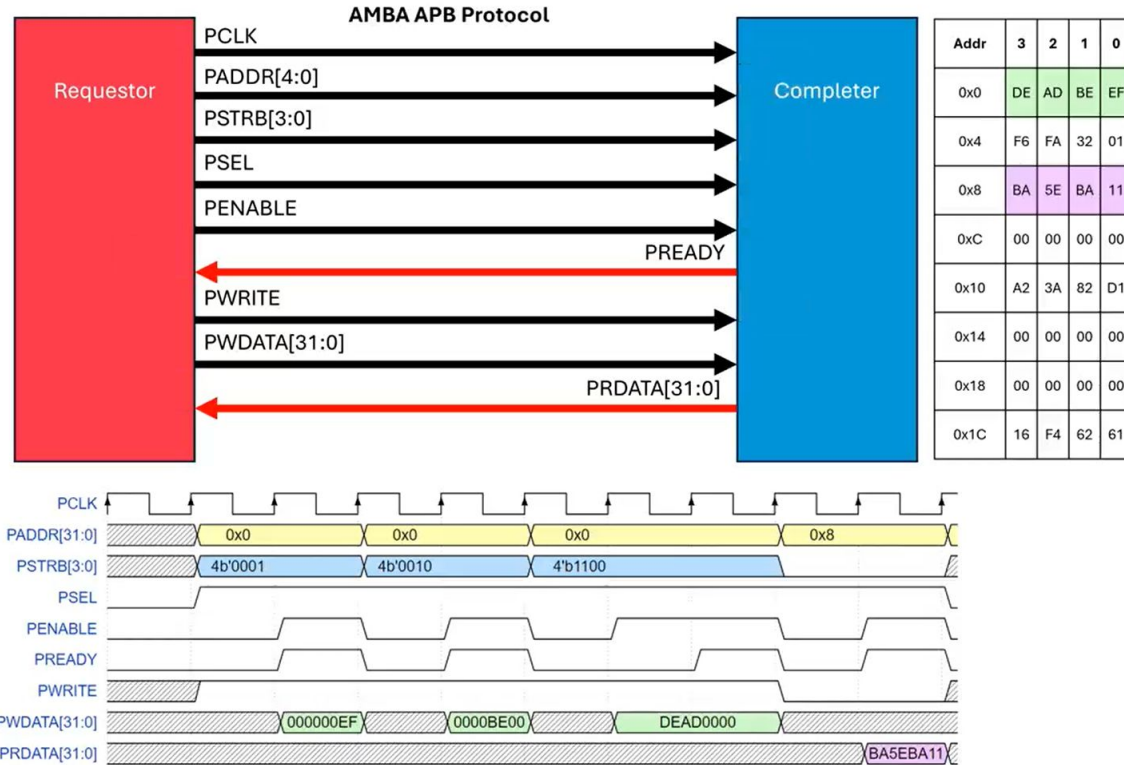
- Entered when a transfer is required.
- The select signal (PSELx) is asserted during this state.
- This state lasts for only one clock cycle.

ACCESS:

- The enable signal (PENABLE) is asserted.
- The address, write signal, select signal, and write data must remain stable during this phase.
- **Exit Conditions:**
 - **Stay in ACCESS:** If PREADY is Low (L).
 - **Go to IDLE:** If PREADY is High (H) and there is no more data to transfer.
 - **Go to SETUP:** If PREADY is High (H) and there is more data pending.



Time Diagram



RX-Unpacker

Unpacker

The CSI-2 RX Unpacker is the first processing stage in the CSI-2 Receiver Data Path. It receives a raw, valid byte stream from the CSI-2 Depacketizer (D-PHY interface) and reconstructs the original parallel pixel data.

This block operates entirely on the high-speed `i_byte_clk`. It supports dynamic "Reverse Gearbox" operations to unpack multi-byte pixel formats (like RGB888 and RAW10) back into a native pixel bus. Its output is designed to be fed directly into an Asynchronous FIFO for clock domain crossing.

Unpacker

4.1.1 Parameters Names

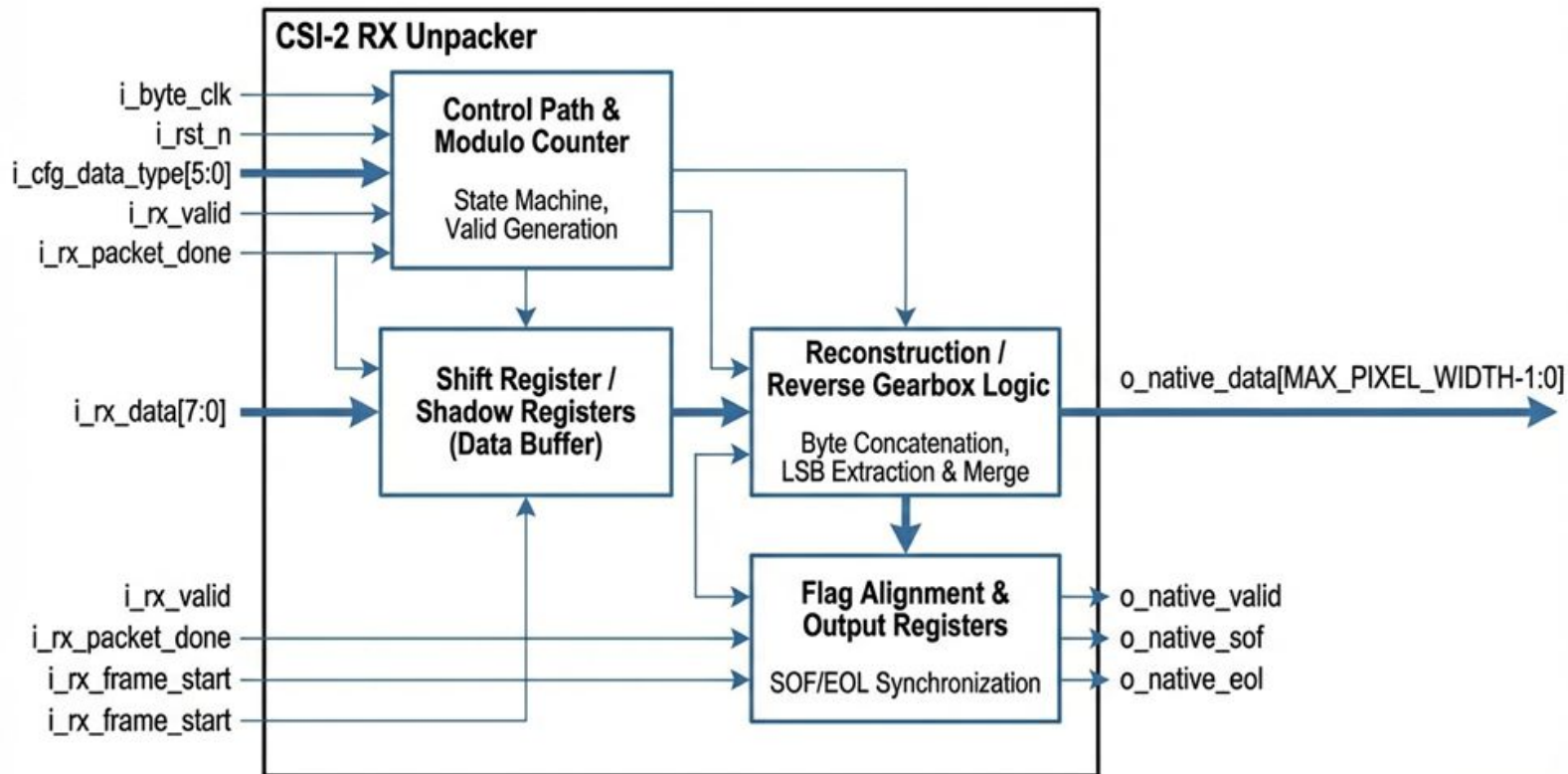
Parameter Name	Default	Description
MAX_PIXEL_WIDTH	24	Maximum width of the reconstructed pixel (e.g., 24 for RGB888, 10 for RAW10).

4.1.2 Ports Names

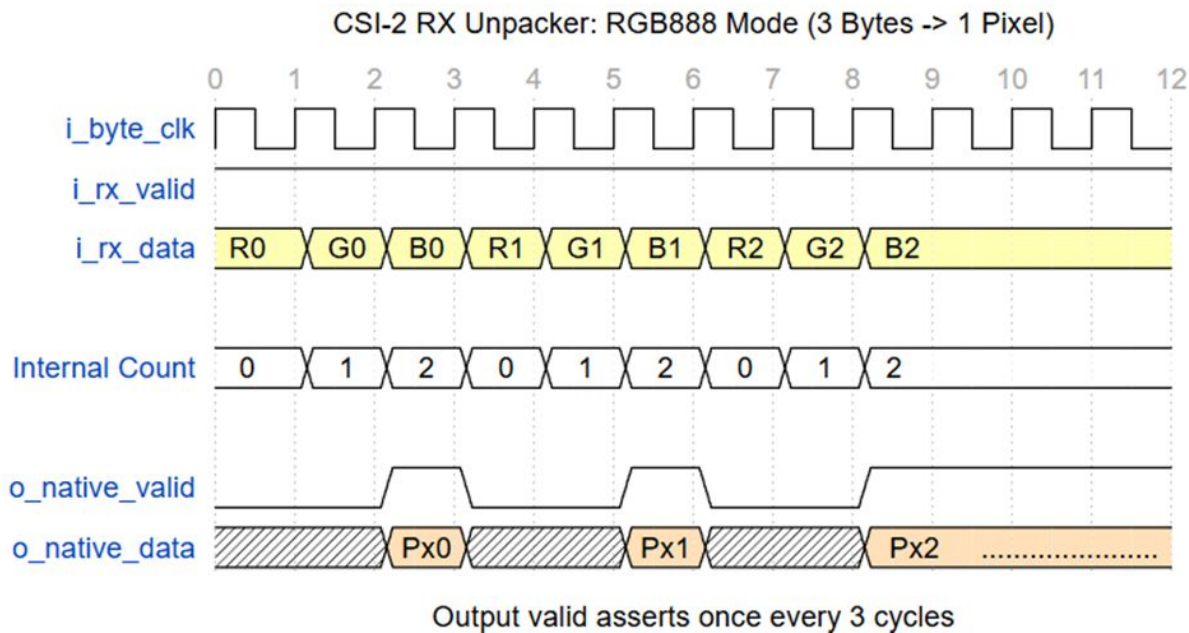
Port Name	Port Width	Port Type	Description
Clocks and Resets			
i_byte_clk	1	Input	High-speed MIPI Byte Clock.
i_rst_n	1	Input	Active-low asynchronous reset.
Configuration			
i_cfg_data_type	6	Input	MIPI Data Type (DT) code. Configures the unpacking ratio.
Byte Input Interface		(From Depacketizer)	
i_rx_data	8	Input	Incoming payload byte stream.
i_rx_valid	1	Input	Valid flag for input byte.
i_rx_packet_done	1	Input	Indicates the last byte of the current line (Payload End).
i_rx_frame_start	1	Input	Single-cycle pulse indicating Start of Frame (from FS Packet).
Native Output Interface		(To Async FIFO)	
o_native_data	WIDTH	Output	Reconstructed Parallel Pixel Data.
o_native_valid	1	Output	Valid flag for the output pixel.
o_native_sof	1	Output	Start of Frame marker (aligned with the first valid pixel of the frame).
o_native_eol	1	Output	End of Line marker (aligned with the last valid pixel of the line).

Unpacker

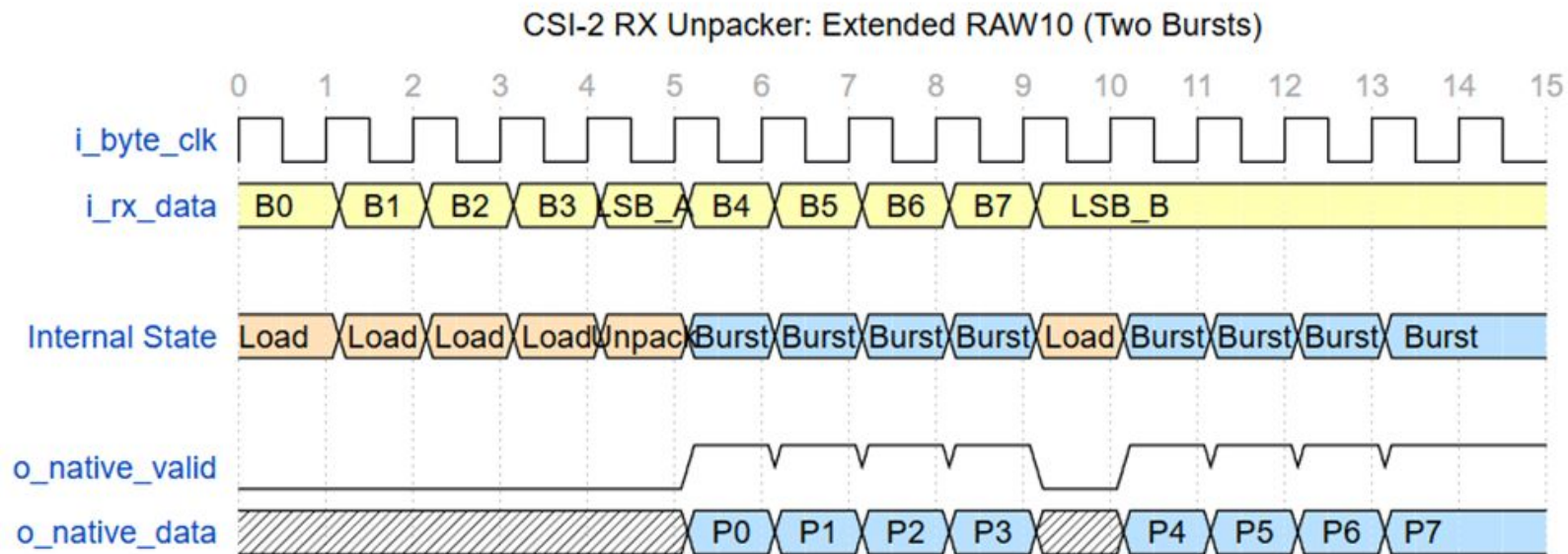
CSI-2 RX Unpacker Block Diagram



Unpacker



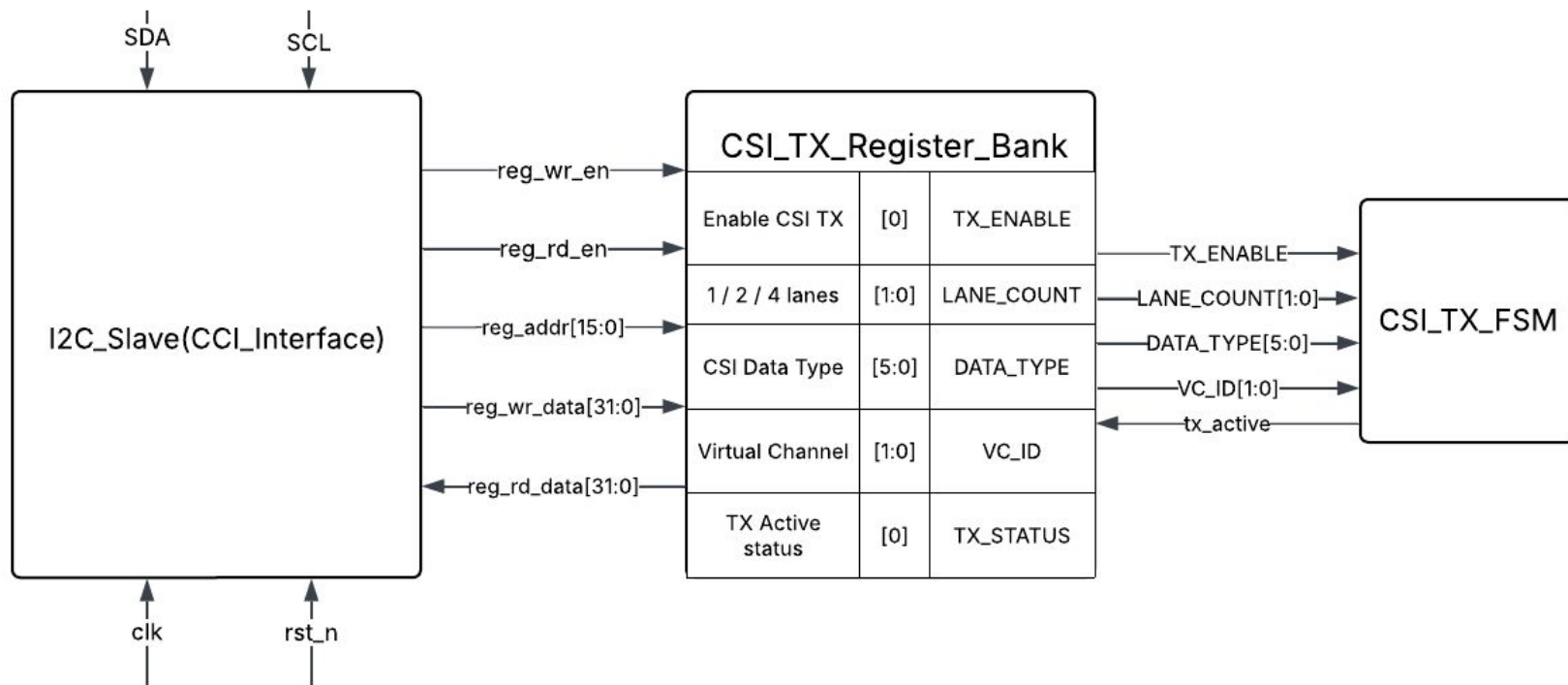
Unpacker



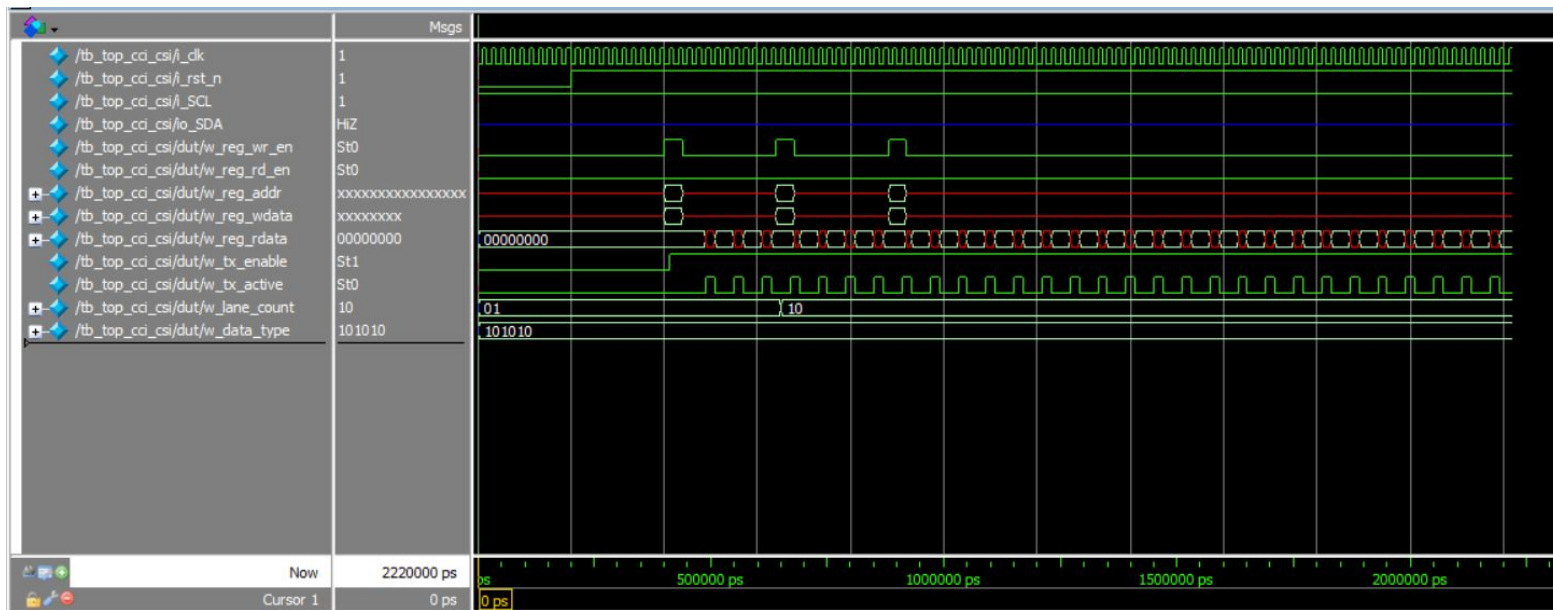
Notice the single 'Load' gap between P3 and P4 due to the 5:4 ratio.

I2C Slave

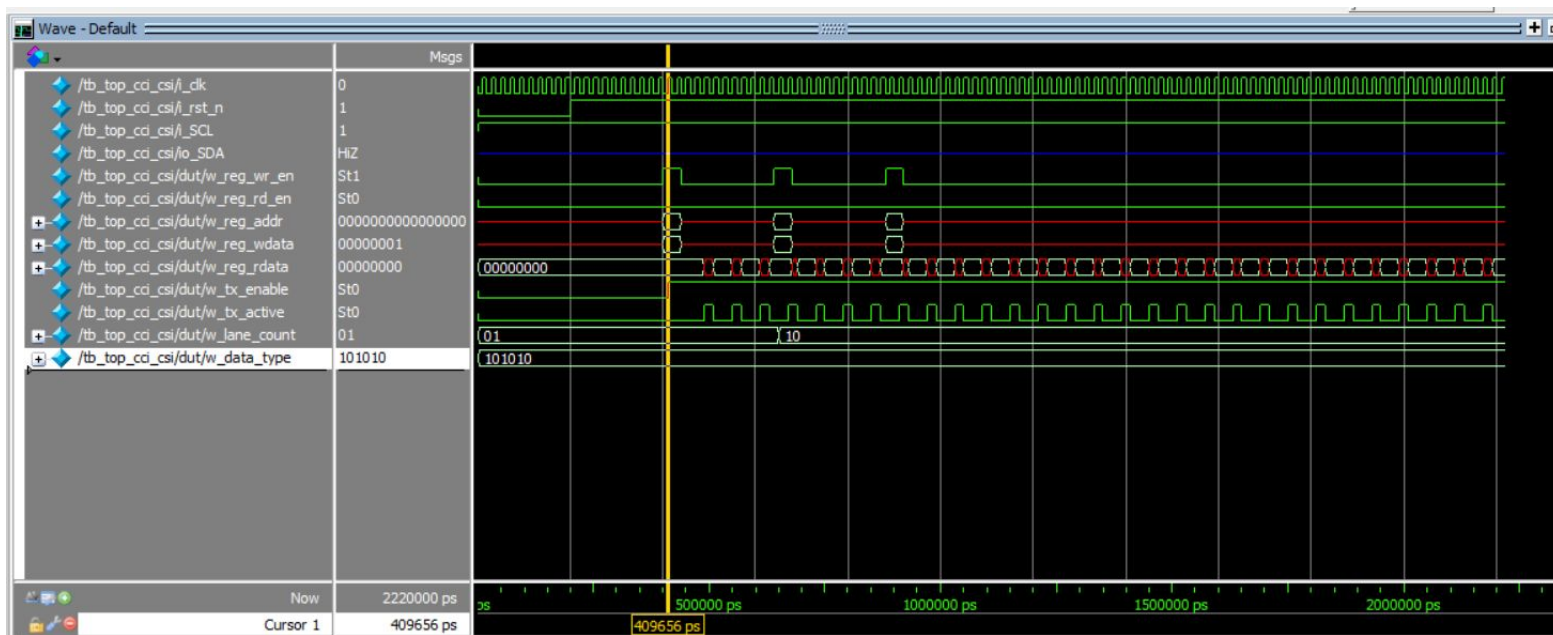
Architecture



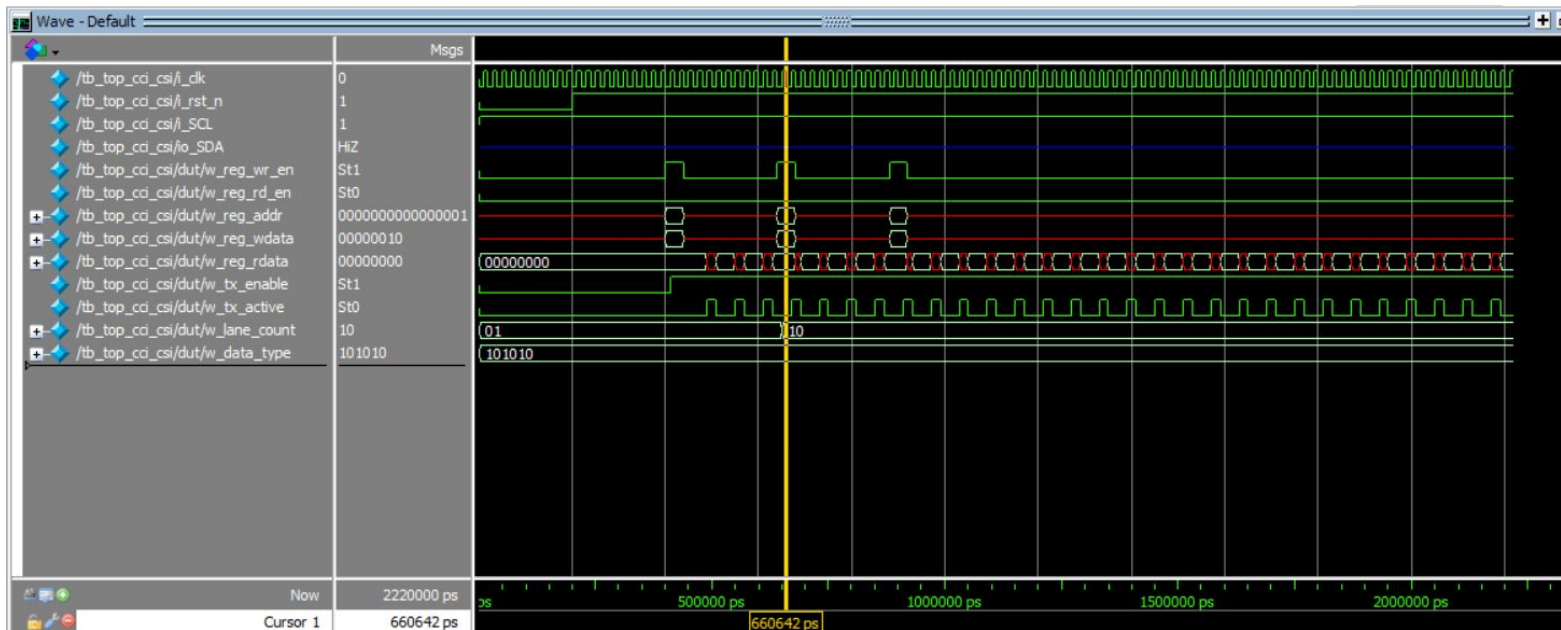
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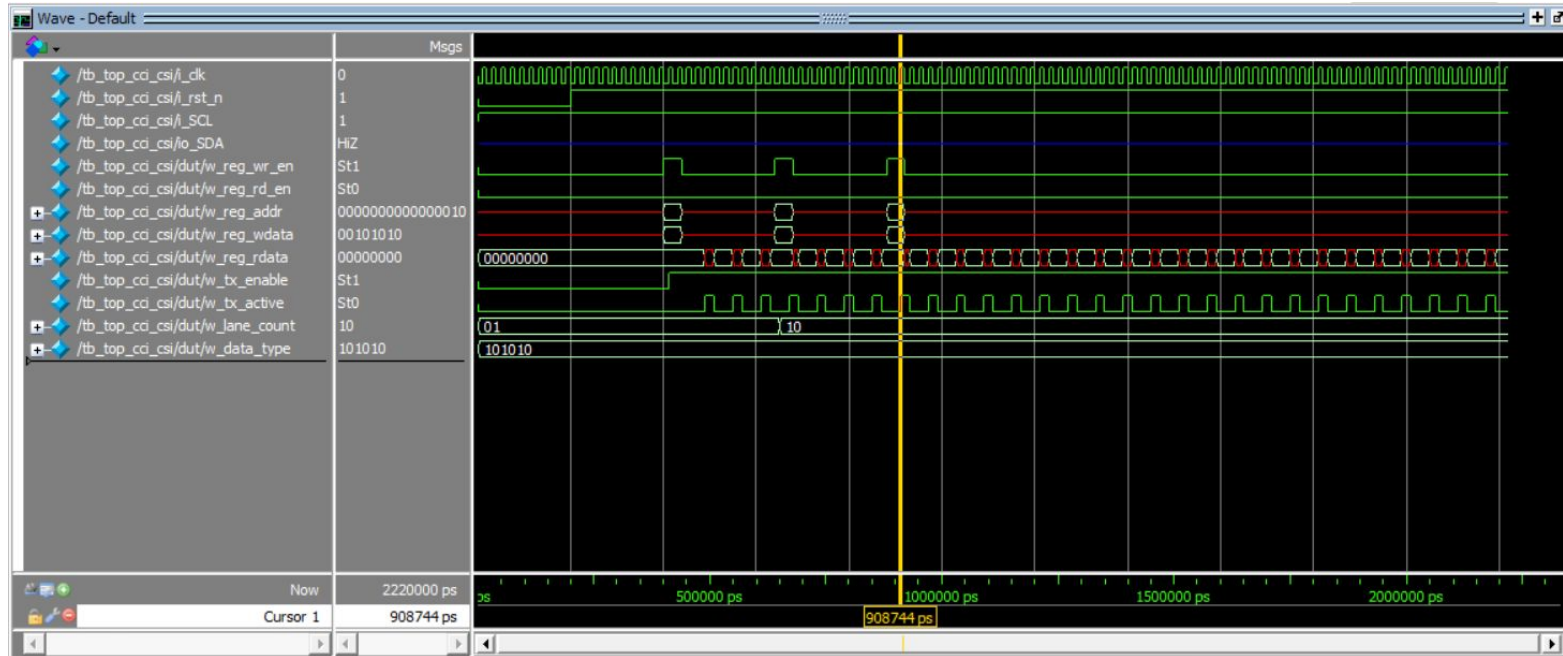
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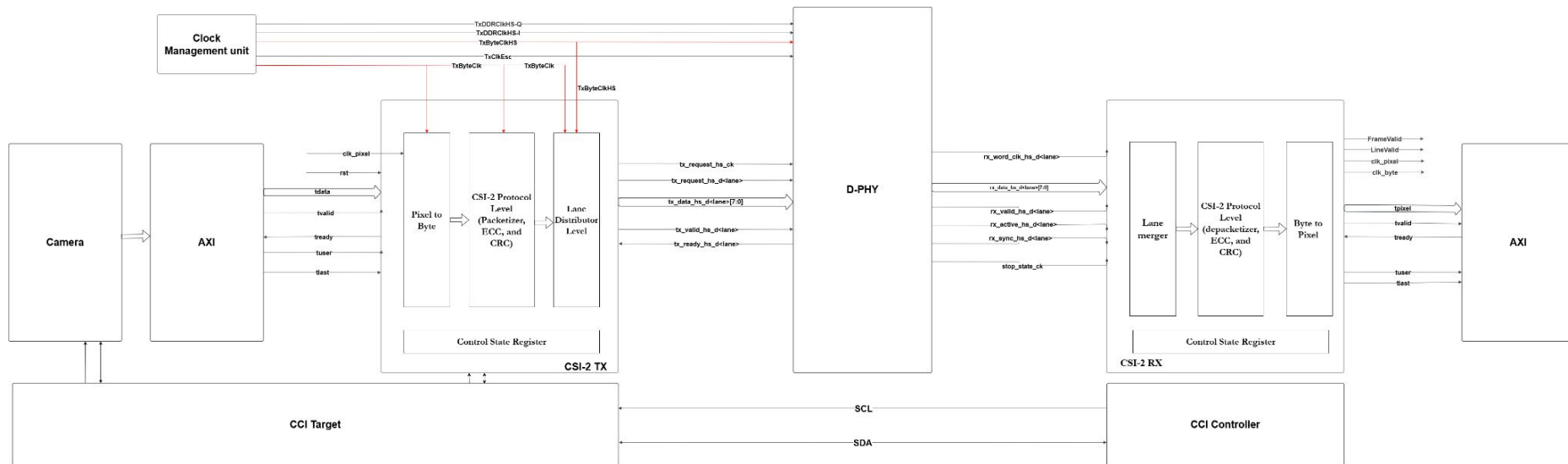
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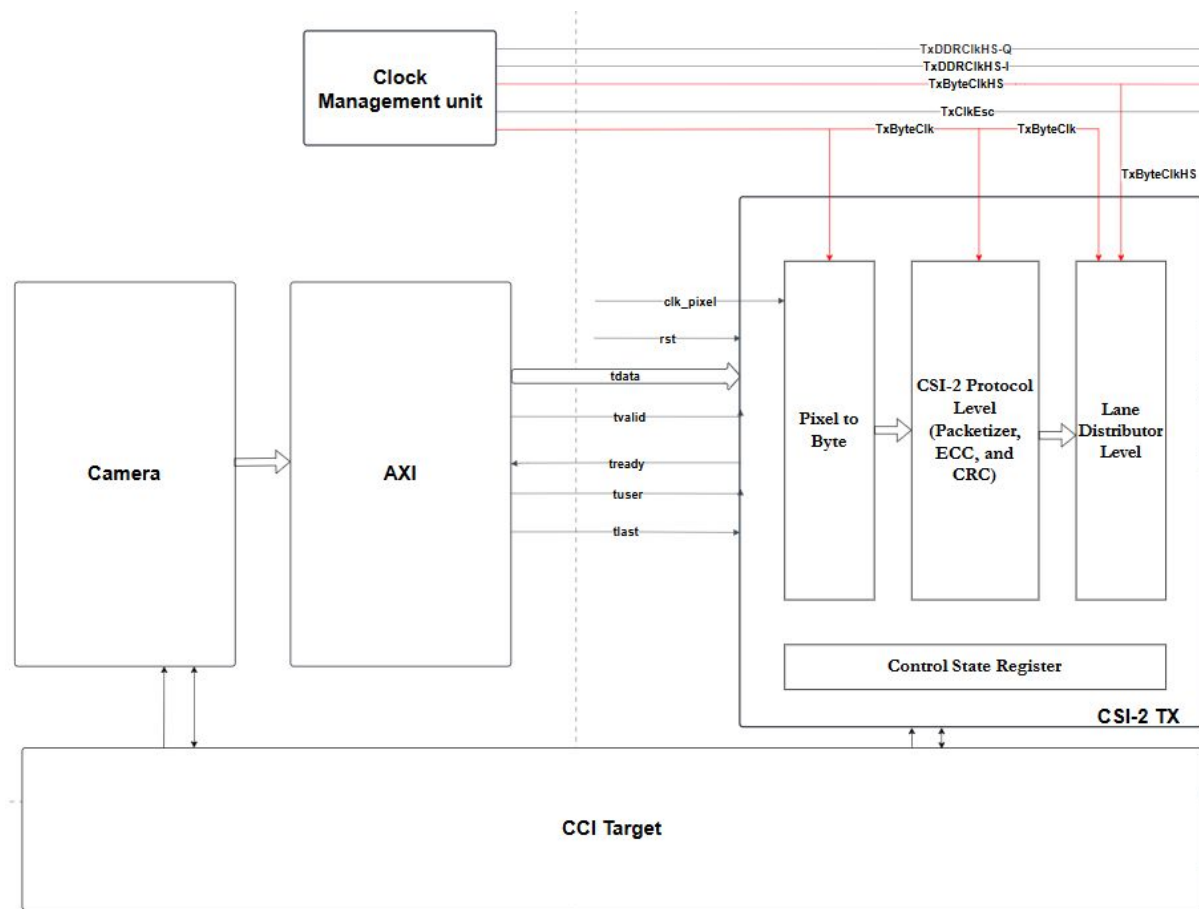


Architecture

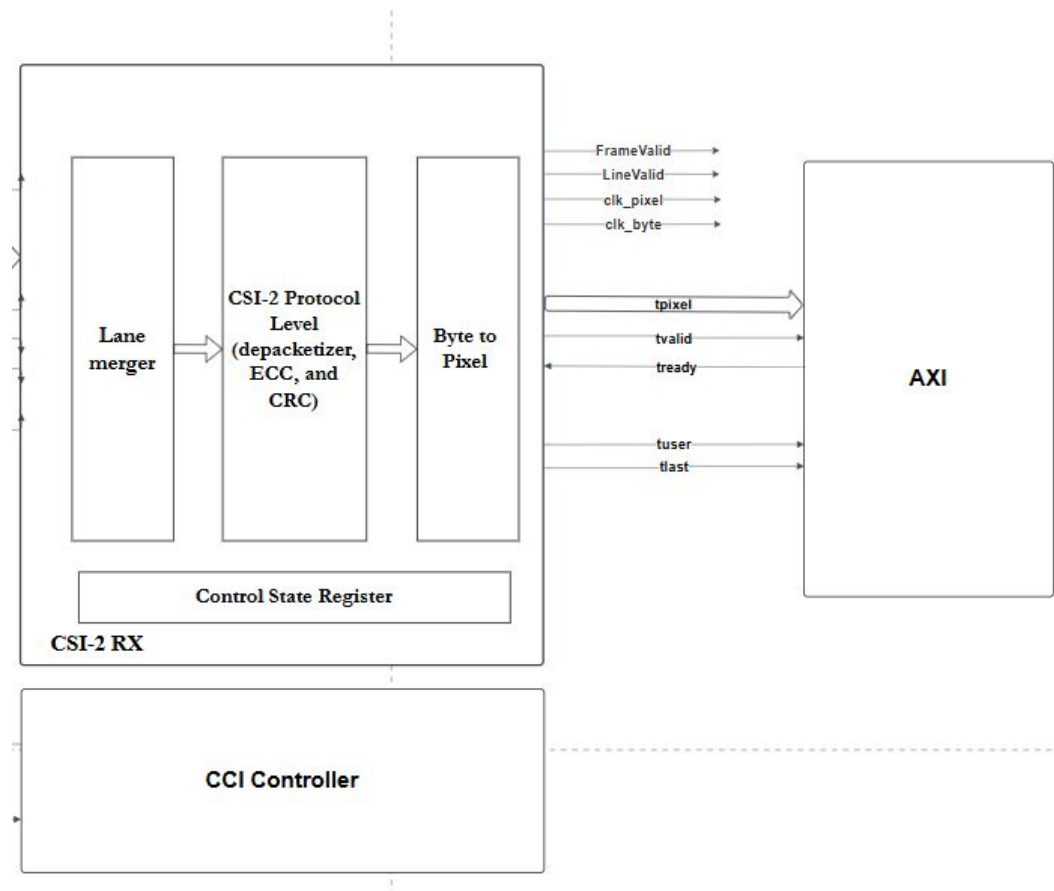
Top-view



Top-view



Top-view



References

References

- [1] MIPI Alliance, "Specification for Camera Serial Interface 2 (CSI-2®) Version 4.0.1," MIPI Alliance, May 23, 2022
- [2] MIPI Alliance, "Specification for D-PHY Version 2.5," MIPI Alliance, Jul. 5, 2019. [Online].
- [3] Arm Limited, *AMBA 3 AHB-Lite Protocol Specification*, IHI 0033E, 2021. [Online]. Available: <https://developer.arm.com/documentation/ihi0024/e/>
- [4] Intel Corporation, "*MIPI CSI-2 Intel® FPGA IP User Guide*," Doc. No. 813926, Version 1.1.0, Apr. 26, 2024. Online. Available: <https://www.intel.com/content/www/us/en/docs/programmable/813926.html>
- [5] Gowin MIPI Pixel to Byte Converter IP
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Thank you!